CAD Design Project 3 – Standard Cell Detailed Placement
Due: 23:59, Nov. 4, 2026

In the standard cell IC design flow, placement is the stage that comes after logic synthesis and before routing. During this step, logic gates and other modules are assigned to specific locations on the chip. Good placement is crucial for improving circuit performance, routing feasibility, thermal distribution, and power consumption. The placement process typically includes two major steps: (1) Global placement generates a rough layout where gate positions may overlap. (2) Detailed placement first legalizes the global placement to ensure that all cells are on valid rows and non-overlapping, and then optimizes wire length while preserving legality. In this project, you are required to implement a detailed placement tool in two stages: **Stage 1** performs legalization to minimize total and maximum cell displacement, and **Stage 2** uses simulated annealing (SA) to minimize wire length while preserving legality. Your program will be evaluated in a Linux environment (e.g., WSL) according to the following requirements.

1. Read a global placement result based on the GSRC Bookshelf file format.
2. For simplicity, there are no macro blocks, and all logic gates have the same row height. A legal placement places every movable cell on a row, aligned to placement sites, entirely within a valid subrow, and without cell overlap. Keep all cell orientations and fixed terminal locations unchanged in both stages.
3. **Stage 1**: Construct a legal placement and minimize both total cell displacement and maximum cell displacement from the input global placement. Save it as OUTPUT_FILE_legal.pl in GSRC Bookshelf format for **Stage 2** to start with.
4. The displacement of cell *i* is derived in Manhattan distance, where *gp* indicates the X/Y coordinates of global placement and *legal* indicates the X/Y coordinates after legalization
$$d_i = \left|x_i^{gp} - x_i^{legal}\right| + \left|y_i^{gp} - y_i^{legal}\right|$$
5. The total displacement is the summation of all cell displacements, where *G* is the set of all logic gates.
$$D_{total} = \sum_{i \in G} d_i$$
6. The maximum cell displacement is the maximum displacement distance among all logic gates.
$$D_{max} = \underset{i \in G}{\text{Max}} \left(d_i\right)$$
7. **Stage 2**: Starting from the **Stage 1** result, implement your own SA algorithm to minimize total half-perimeter wire length (HPWL). Maintain legality for all logic gates and output the best placement found as OUTPUT_FILE.pl in GSRC Bookshelf format.
8. For each net, HPWL is the bounding box width plus height of the corresponding cell centers: HPWL(net) = (max x − min x) + (max y − min y). For simplicity, ignore pin offsets and

assume all pins are at the cell center. The total HPWL is the unweighted sum over all nets.

9. Print total displacement, maximum displacement, and total HPWL for **Stage 1**, and only total HPWL for **Stage 2**, to standard output.
10. Submit a research report as a single PDF of at most 3 pages, including figures and tables. Explain initial legal solution construction and subsequent SA optimization while preserving legality. Support your strategies with analysis, experimental evidence, and trade-offs. Report total and maximum displacement for **Stage 1**, and compare total HPWL before and after SA on the same inputs. Also, plot the *current* and *best-so-far* wirelength against the number of attempted SA iterations, and justify the design, effectiveness, and observed behaviors of your simulated annealing algorithm. A GUI, auxiliary analysis tools, or your own special features are highly encouraged.
11. Upload a tarball (*.tgz) of your C/C++ program source code and a text ReadMe file, which illustrates how your program is to be compiled and executed in a Linux environment. (NOTE: The uploaded file name should be the same as your student ID.)

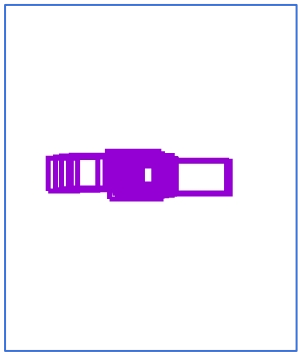
Reference

[1] http://vlsicad.eecs.umich.edu/BK/ISPD06bench/BookshelfFormat.txt

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SYNOPSIS for Detailed Placer
%> dp INPUT_FILE OUTPUT_FILE

Run-time Examples:

%> ls toy.*
toy.aux  toy.nets  toy.nodes
toy.pl   toy.scl   toy.wts
```



```
%> dp toy output
Total displacement (legal): 62531.2
Maximum displacement (legal): 5219.7
Total HPWL (legal): 757296
Total HPWL (SA): 690185

%> ls output*
output.aux  output.nets     output.nodes
output.pl   output_legal.pl output.scl
output.wts
```

